

ABSTRAK

Dewasa ini, transformasi digital melalui layanan *e-government* kian masif dikembangkan untuk meningkatkan efektivitas pelayanan publik, tetapi layanan pendaftaran nikah dan cerai masih menghadapi tantangan pada kompleksitas administratif dan alur proses yang belum terstandarisasi. Penelitian ini bertujuan untuk mengembangkan prototipe sistem pendaftaran nikah dan cerai dengan menerapkan BPMN *engine* untuk mendukung otomatisasi proses bisnis. Pengembangan sistem dilakukan menggunakan pendekatan *Hybrid Iterative Prototyping Model* (HIPM) yang mengintegrasikan prinsip *Software Development Life Cycle* (SDLC), *Design Thinking*, dan *Business Process Management* (BPM) melalui tiga iterasi pengembangan pada sisi server. Hasil pengujian fungsionalitasnya menunjukkan indeks keberhasilan mencapai 96,07% untuk sistem pendaftaran nikah dan 96,42% untuk pendaftaran cerai. Dengan begitu, prototipe berhasil melampaui target yang ditetapkan, yakni $\geq 85\%$. Evaluasi kinerja proses bisnis menggunakan *Key Performance Indicator* (KPI) menunjukkan hasil yang berbeda signifikan antara kedua sistem. Sistem pendaftaran nikah berhasil memberikan optimasi tinggi dengan indeks efisiensi sebesar 70,42% dan indeks capaian target sebesar 161,3%; menunjukkan bahwa sistem tidak hanya mampu mengurangi beban proses bisnis secara besar dibandingkan kondisi awal (*as-is*), tetapi juga berhasil melampaui target yang diharapkan. Hasil berbeda didapatkan oleh sistem pendaftaran cerai, optimasi yang dihasilkan relatif rendah dengan indeks efisiensi sebesar 15% dan indeks capaian target sebesar 70,99%; menunjukkan bahwa perubahan beban proses bisnis yang terjadi relatif kecil dan target kinerja yang diharapkan setelah penerapan sistem tidak tercapai. Penelitian ini juga menemukan bahwa otomatisasi proses bisnis menggunakan BPMN *engine* paling efektif digunakan untuk mengoptimalkan proses bisnis dengan kondisi awal bersifat tidak terstruktur, khususnya pada aktivitas administratif yang dapat distandarkan melalui aturan proses yang eksplisit. Akan tetapi, efektivitas tersebut dibatasi oleh adanya tahapan proses yang secara inheren bergantung pada regulasi pihak eksternal dan keputusan non-sistem. Dengan demikian, penelitian ini memberikan kontribusi dalam memetakan kapabilitas otomatisasi pada layanan publik yang memiliki kompleksitas dan karakteristik birokrasi yang berbeda.

Kata kunci— Optimasi Proses Bisnis, BPMN *engine*, Hybrid Iterative Prototyping Model (HIPM), Key Performance Indicator (KPI), Sistem Automasi.

ABSTRACT

Nowadays, digital transformation through e-government services is increasingly being developed to improve the effectiveness of public services; however, marriage and divorce registration services still face challenges related to administrative complexity and non-standardized process flows. This study aims to develop a prototype of a marriage and divorce registration system by applying a BPMN engine to support business process automation. The system development was carried out using the Hybrid Iterative Prototyping Model (HIPM) approach, which integrates the principles of the Software Development Life Cycle (SDLC), Design Thinking, and Business Process Management (BPM) through three development iterations on the server side. The functional testing results show a success index of 96.07% for the marriage registration system and 96.42% for the divorce registration system. Thus, the prototype successfully exceeded the predetermined target of $\geq 85\%$. The evaluation of business process performance using Key Performance Indicators (KPI) shows significantly different results between the two systems. The marriage registration system achieved high optimization with an efficiency index of 70.42% and a target achievement index of 161.3%, indicating that the system was not only able to substantially reduce the business process workload compared to the initial condition (as-is), but also successfully exceeded the expected targets. Different results were obtained for the divorce registration system, where the resulting optimization was relatively low with an efficiency index of 15% and a target achievement index of 70.99%, indicating that the change in business process workload was relatively small and the expected performance targets after system implementation were not achieved. This study also finds that business process automation using a BPMN engine is most effective for optimizing business processes with initially unstructured conditions, particularly administrative activities that can be standardized through explicit process rules. However, this effectiveness is limited by the presence of process stages that inherently depend on external regulatory authorities and non-system decisions. Thus, this study contributes to mapping the capabilities of automation in public services that have different levels of bureaucratic complexity and characteristics.

Keywords— Business Process Optimization, BPMN engine, Hybrid Iterative Prototyping Model (HIPM), Key Performance Indicator (KPI), Automation System.